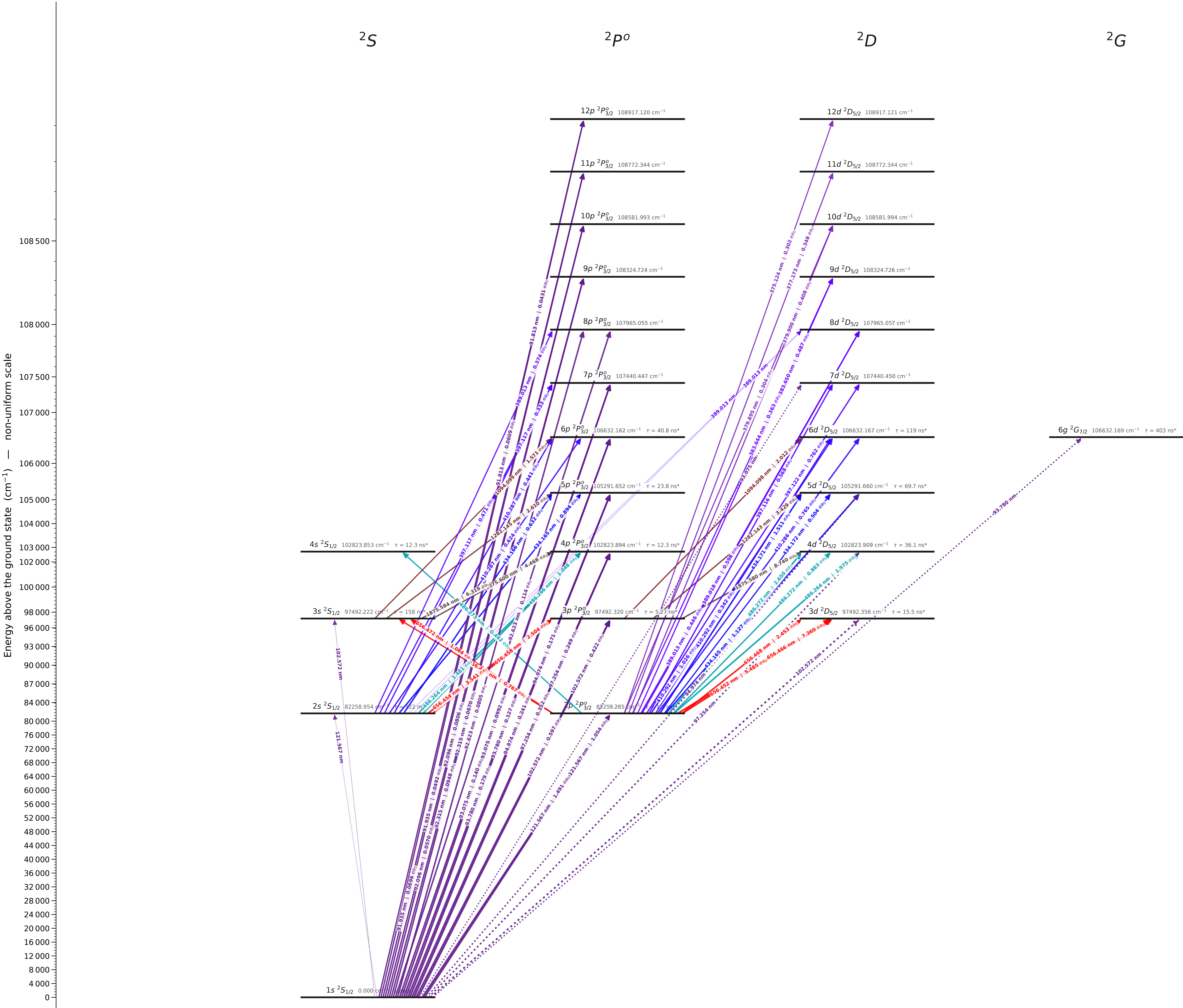
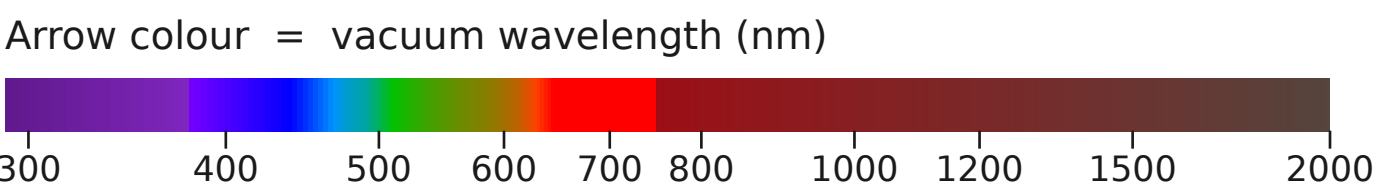


Hydrogen

H (H I). Strongest classified transitions below 2 μm (NIST): 45 levels, 74 lines.

H⁺ ionisation limit 109678.77 cm⁻¹ = 13.59843 eV (91.175 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e r J' \rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
1s ² S _{1/2} – 3p ² P _{1/2} ^o	102.5723	–	2922.743279	0.422	1.672e+08	30.19	88.2	1S-3S two-photon transition
1s ² S _{1/2} – 3s ² S _{1/2}	102.5723	–	2922.743279	–	–	1.005	–	1S-3S two-photon transition
1s ² S _{1/2} – 2p ² P _{1/2} ^o	121.5668	–	2466.0713	1.491	6.265e+08	99.71	100	Lyman-alpha, 121.6 nm: Doppler laser cooling of magnetica...
1s ² S _{1/2} – 2s ² S _{1/2}	121.5673	–	2466.061413	–	8.229e+00	1.31e-06	100	1S-2S two-photon transition
1s ² S _{1/2} – 2p ² P _{1/2} ^o	121.5674	–	2466.0604	1.054	6.265e+08	99.71	100	Lyman-alpha, 121.6 nm: Doppler laser cooling of magnetica...
2s ² S _{1/2} – 8d ² D _{5/2}	389.0127	388.9025	770.649562	–	–	–	–	2S-8D two-photon transition
2p ² P _{1/2} ^o – 8d ² D _{3/2}	389.0127	388.9025	770.649504	0.446	1.710e+06	–	–	2S-8D two-photon transition
2s ² S _{1/2} – 8p ² P _{3/2} ^o	389.0127	388.9025	770.649504	0.374	1.202e+06	–	–	2S-8D two-photon transition
2s ² S _{1/2} – 8d ² D _{3/2}	389.0127	388.9025	770.649504	–	–	–	–	2S-8D two-photon transition

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹H hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
1s ² S _{1/2}	1420.405752	–	F=0: -1065.30 F=1: +355.10
2p ² P _{1/2} ^o	59.1695(6)	–	F=0: -44.38 F=1: +14.79
2s ² S _{1/2}	177.556834(7)	–	F=0: -133.17 F=1: +44.39
2p ² P _{3/2} ^o	11.82580(12)	–	F=1: -14.78 F=2: +8.87
3p ² P _{1/2} ^o	17.53180(18)	–	F=0: -13.15 F=1: +4.38
3s ² S _{1/2}	52.609473	–	F=0: -39.46 F=1: +13.15
3d ² D _{3/2}	2.1025(20)	–	F=1: -2.63 F=2: +1.58
3p ² P _{3/2} ^o	3.50396(3)	–	F=1: -4.38 F=2: +2.63
3d ² D _{5/2}	0.9003(9)	–	F=2: -1.58 F=3: +1.12
4p ² P _{1/2} ^o	7.39623(7)	–	F=0: -5.55 F=1: +1.85
4s ² S _{1/2}	7.39623(7)	–	F=0: -5.55 F=1: +1.85
4d ² D _{3/2}	0.8870(9)	–	F=1: -1.11 F=2: +0.67
4p ² P _{3/2} ^o	1.478235(15)	–	F=1: -1.85 F=2: +1.11
4d ² D _{5/2}	0.3798(4)	–	F=2: -0.67 F=3: +0.47
5p ² P _{1/2} ^o	3.78586(4)	–	F=0: -2.84 F=1: +0.95
5d ² D _{3/2}	0.4541(4)	–	F=1: -0.57 F=2: +0.34
5p ² P _{3/2} ^o	0.756857(8)	–	F=1: -0.95 F=2: +0.57
5d ² D _{5/2}	0.19447(20)	–	F=2: -0.34 F=3: +0.24
6p ² P _{1/2} ^o	2.191470(22)	–	F=0: -1.64 F=1: +0.55
6d ² D _{3/2}	0.26280(25)	–	F=1: -0.33 F=2: +0.20
6p ² P _{3/2} ^o	0.437996(4)	–	F=1: -0.55 F=2: +0.33
6d ² D _{5/2}	0.11253(10)	–	F=2: -0.20 F=3: +0.14
7p ² P _{1/2} ^o	1.380050(14)	–	F=0: -1.03 F=1: +0.34
7d ² D _{3/2}	0.16550(15)	–	F=1: -0.21 F=2: +0.12
7p ² P _{3/2} ^o	0.275823(3)	–	F=1: -0.34 F=2: +0.21
7d ² D _{5/2}	0.07087(7)	–	F=2: -0.12 F=3: +0.09
8p ² P _{1/2} ^o	0.924525(9)	–	F=0: -0.69 F=1: +0.23
8d ² D _{3/2}	0.11088(11)	–	F=1: -0.14 F=2: +0.08
8p ² P _{3/2} ^o	0.184780	–	F=1: -0.23 F=2: +0.14
8d ² D _{5/2}	0.04748(5)	–	F=2: -0.08 F=3: +0.06
9p ² P _{1/2} ^o	0.649323(6)	–	F=0: -0.49 F=1: +0.16
9d ² D _{3/2}	0.07787(8)	–	F=1: -0.10 F=2: +0.06
9p ² P _{3/2} ^o	0.129776	–	F=1: -0.16 F=2: +0.10
9d ² D _{5/2}	0.03334(3)	–	F=2: -0.06 F=3: +0.04
10p ² P _{1/2} ^o	0.473356(5)	–	F=0: -0.35 F=1: +0.12
10d ² D _{3/2}	0.094607	–	F=1: -0.12 F=2: +0.07
10p ² P _{3/2} ^o	0.094607	–	F=1: -0.12 F=2: +0.07
10d ² D _{5/2}	0.024307(23)	–	F=2: -0.04 F=3: +0.03
11p ² P _{1/2} ^o	0.355639(4)	–	F=0: -0.27 F=1: +0.09
11p ² P _{3/2} ^o	0.071080	–	F=1: -0.09 F=2: +0.05
11d ² D _{5/2}	0.018263(17)	–	F=2: -0.03 F=3: +0.02
12p ² P _{1/2} ^o	0.273933(3)	–	F=0: -0.20 F=1: +0.07
12p ² P _{3/2} ^o	0.054749	–	F=1: -0.07 F=2: +0.04

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
1s ² S _{1/2} – 2s ² S _{1/2}	121.567	2-1	670994.334606(15)
2s ² S _{1/2} – 8d ² D _{5/2}	389.013	2-1	209691.265(9)
2p ² P _{1/2} ^o – 8d ² D _{3/2}	389.013	2-1	209691.252(10)
2s ² S _{1/2} – 8p ² P _{3/2} ^o	389.013	2-1	209691.252(10)
2s ² S _{1/2} – 8d ² D _{3/2}	389.013	2-1	209691.252(10)

Shift = ν(first isotope) – ν(second isotope).